BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 139 · Lesson 139 · SQL INSERT, UPDATE and DELETE Safely

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is the difference between WHERE and ORDER BY in a query?

Q2. What are the outputs of a half adder when A=1 and B=1?

Q3. In Big O notation, what is binary search worst-case complexity?

Q4. When does a linear search stop?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

INSERT — Adds new rows.

UPDATE — Changes values in existing rows.

DELETE — Removes matching rows.

WHERE — Restricts affected rows; omission can affect every row.

New learning

Create rows — INSERT INTO Table(columns) VALUES(...).

Change rows safely — UPDATE Table SET column=value WHERE key=...

Delete deliberately — DELETE FROM Table WHERE key=...; verify matches before writing.

Worked example

A Book table has BookID=8,Stock=4. Write an UPDATE to set Stock=3 for that row.

1. Identify BookID 8 — Find the given data and name the required result.

2. UPDATE Stock = 3 — Explain why this intermediate stage is necessary.

3. WHERE BookID = 8 — UPDATE Book SET Stock = 3 WHERE BookID = 8; Before running, SELECT the matching row. Without WHERE, the stock column of all books would be changed.

We Do · Practise with guidance

Add a new Club record and delete only the club with ClubID=12.

Insert club 24

Find the given data and name the required result.

Confirm club 12 exists

Explain why this intermediate stage is necessary.

DELETE WHERE ClubID=12

INSERT INTO Club (ClubID, Name) VALUES (24, 'Robotics'); DELETE FROM Club WHERE ClubID = 12; Foreign-key dependencies may prevent delete until related records are handled by defined policy.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Write SQL inserting StudentID 22 and Name 'Kai' into Student. [4]

Question 2

A teacher must change a single pupil's YearGroup to 13. Explain a safe SQL statement. [6]

Question 3

Explain risks in DELETE FROM Booking; and suggest protections in a relational production database. [8]

Exit Ticket · Answer before checking

UPDATE Book SET Stock=0; changes exactly one Book record.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. WHERE filters rows; ORDER BY arranges returned rows.

Q2. Sum 0, carry 1.

Q3. O(log n) for sorted random-access data.

Q4. On a match or after examining all items.

You Do pages · Independent questions

Q1. INSERT INTO Student (StudentID, Name) VALUES (22, 'Kai'); [4].

Q2. UPDATE Student SET YearGroup=13 [3] WHERE StudentID=<specific existing ID> [2]; confirm selected row before applying [1].

Q3. Missing WHERE deletes all rows [2], FK/referential impacts [2], transactions/rollback and backups [2], permission controls and dry-run SELECT [2].

Exit Ticket · Model answer

False. With no WHERE it updates every row.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: